



M003 Week 6 — English Worksheet (Advanced)

Topic: Recreate the Ender Dragon (x, y, z) · **Course:** 3D Coordinates · **Level:** Advanced · **Time:** about 45 minutes

A big blocky **ender dragon** stands in your world — black blocks, purple eyes. A big model is a set of **parts**: a long **body** (a fill box stretched along one axis), a **head** at the front, two **wings** that spread out along x, and a **tail** that steps back and down. Everything uses **(x, y, z)** — x **across**, y **up**, z **forward** (deep). Read each part carefully and plan your corners before you build.

● Stand on your **home spot** (red block) every run. Move your feet, move your head.

1 · Predict 🧠

Read each snippet. Do **not** run it. Write your prediction **and the reason**.

```
fill black from (0, 8, 0) to (0, 8, 10)
```

How long is the body, and which way does it run? Why?


```
fill black from (1, 8, 5) to (5, 8, 5)
fill black from (-5, 8, 5) to (-1, 8, 5)
```

Two wings. How wide is each one? Are they the same size on both sides? Why?


```
place black block at (0, 7, -1)
place black block at (0, 6, -2)
place black block at (0, 5, -3)
```

This is the tail behind the body. As z gets more negative, what happens to the height? Why does that make it droop?


```
fill black from (0, 8, 0) to (0, 8, 10)
place black block at (0, 9, 11)
```

Body, then head. Is the head in front of or behind the body? Is it higher or lower? Why?

2 · Spot the Bug 🐛

Bug A — These should be **two matching wings**, spreading the same distance out on each side from the body at $x = 0$. Right now the left wing is much shorter than the right.

```
fill black from (1, 8, 5) to (5, 8, 5)
fill black from (-2, 8, 5) to (-1, 8, 5)
```

Write the fixed code:

Why was it wrong? Why does your fix work? 2–3 sentences.

Bug B — This should be a **long body** stretched **deep along z**, from $(0, 8, 0)$ to $(0, 8, 10)$. Right now it lies flat across the ground along x instead.

```
fill black from (0, 8, 0) to (10, 8, 0)
```

Write the fixed code:

Why was it wrong? Why does your fix work? 2–3 sentences.

Bug C — The **tail** should step **back and down** behind the body: each block goes one deeper and one lower. Right now every tail block sits at the same height, so the tail floats straight out instead of drooping.

```
place black block at (0, 8, -1)
place black block at (0, 8, -2)
place black block at (0, 8, -3)
```

Hint: look at the **y** numbers. A tail that droops must lower y each step, not keep it the same.

Write the fixed code:

Why was it wrong? Why does your fix work? 2–3 sentences.

3 · Build It 📷

Look at the dragon in your world. Walk around it. **Plan your corners first**, then copy it part by part on your home spot:

```
fill black from (0, 8, 0) to (0, 8, 10)
place black block at (0, 9, 11)
fill black from (1, 8, 5) to (5, 8, 5)
fill black from (-5, 8, 5) to (-1, 8, 5)
place black block at (0, 7, -1)
place black block at (0, 6, -2)
place black block at (0, 5, -3)
```

That is body, head, right wing, left wing, and a stepped tail. Add two **purple** eye blocks on the head. Keep the wings the same length on both sides.

MODIFY challenge: change one thing that is your own idea — a new colour, an extra part, or a bigger dragon. Write down what you changed.

Send a photo or video. Then finish these — 5 to 6 sentences.

I built the dragon as these parts: ...

My body runs along the ... axis, from (... , ... , ...) to (... , ... , ...).

To make the wings match, I ...

My tail steps back and down by changing the ... number and the ... number each time.

For my MODIFY challenge, I changed ...

One tricky moment was when ...

I built the dragon as these parts: ...

My body runs along the ... axis, from (... , ... , ...) to (... , ... , ...).

To make the wings match, I ...

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For my MODIFY challenge, I changed ...

One tricky moment was when ...

4 · Explain It 🎥

Film yourself showing your dragon. Use these words: **x, y, z, body, head, wing, tail.**

1. Show your home spot. Why stand on it every run?
2. Point at the body. Say which axis it runs along and why.
3. Point at both wings. Show they match, and **say in your own words** how you kept them the same.
4. Point at the tail. Say which numbers change to make it step back and down.

Write a short paragraph (5–6 sentences) explaining how you built your dragon and how the (x, y, z) numbers control each part:

Last question — what would you teach a friend first if they wanted to build this dragon? Why that part?

Submit ✓

Send this worksheet + your walkthrough video to teacher on KakaoTalk.